LinkedIn edunet foundation

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Pay-Lens: visualizing salary truth with AI & BI

Project Domain

[Excel / Power BI with Copilot & chatbot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

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ABSTRACT OF THE PROJECT

This project, undertaken as part of the AICW Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP, addresses critical challenges in payroll management and compensation transparency. Payroll data is often raw, inconsistent, and locked within complex systems, creating information asymmetry between employers and employees. Workers, particularly gig and blue-collar employees, frequently lack clarity on how their wages are calculated, while organizations face compliance risks related to statutory requirements such as the mandated 2× overtime pay under Indian labour laws.

The objectives of this project are to collect and preprocess a payroll dataset of 1,000 records, apply ratio analysis against market benchmarks, and build an interactive Power BI dashboard to visualize salary structures, pay equity, and compliance. Using Excel formulas, Power Query, and DAX measures, the project standardizes compensation data and calculates indices such as the Market Positioning Index. AI tools, including Microsoft Copilot, can be integrated to generate automated discrepancy alerts, ensuring that overtime payments and statutory deductions are correctly applied.

The methodology follows a structured five-phase workflow: dataset collection, data cleaning, analytical modelling, dashboard visualization, and documentation. The resulting system delivers automated verification of payroll compliance, enhances financial literacy for employees through clear deduction breakdowns, and equips HR professionals with AI-assisted analytics capabilities.

Key outcomes include the development of an audit-ready payroll dataset, a decision-intelligence dashboard highlighting pay equity and attrition risks, and an AI-augmented framework for anomaly detection. By bridging transparency, compliance, and analytical gaps, this project demonstrates how data analytics and AI can transform payroll systems into tools of fairness, accountability, and organizational trust.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project.

1.1 Problem Statement

- **What is the main problem or challenge in your chosen domain?**

  Payroll management systems often contain raw, inconsistent, and poorly structured data. Employees, especially gig and blue-collar workers, face information asymmetry because they cannot clearly see how their wages are calculated from Gross CTC to net salary. Employers, on the other hand, struggle with ensuring compliance to statutory requirements such as the mandated 2× overtime pay. Disputes also arise when employees claim more overtime hours than employers record, and when updated payroll policies are not communicated effectively, leaving employees unaware of their entitlements.

- **Why is this problem significant, and what is its real-world impact?**

  This problem is critical because it involves Legal Compliance and Financial Stability. Under Indian labour laws (such as the Factories Act), overtime must be paid at 2x the ordinary rate. In the real world, poor transparency often leads to systemic underpayment of overtime or incorrect statutory deductions (PF, Professional Tax) that go unnoticed by the workforce. Furthermore, pay inequity—where similar roles have widely different packages—erodes organizational trust and fairness.

- **Who is affected by this problem — what are the target users or organizations?**
  - **Employees:** Gig workers, blue-collar staff, and interns are most vulnerable to wage loss and policy miscommunication.
  - **HR Departments:** Struggle with compliance verification, pay equity analysis, and high turnover costs.
  - **Organizations:** Face financial penalties, reduced productivity, and reputational risks due to payroll disputes.

- **What are the limitations of existing approaches or tools currently available?**
  - Traditional payroll systems focus on calculation but lack transparency for employees.
  - Manual attendance systems and outdated reporting tools fail to resolve overtime disputes.
  - Benchmarking against market salaries is often outsourced and not automated internally.

- Policy updates are communicated through circulars or emails, which employees may forget or overlook.
- AI tools are available but underutilized due to lack of prompt engineering skills among HR professionals.

1.2 Motivation

As a student from the finance domain, I was motivated to choose this project because payroll management is not only a financial function but also a compliance-critical area that directly impacts organizational trust. Payroll involves calculations of gross CTC, statutory deductions, allowances, and overtime payments — all of which require accuracy, transparency, and adherence to labour laws.

Along with this I also studied cases around this domain where A Bengaluru-based IT professional sued her employer for unpaid overtime. The employer was held guilty under section 2(iii) of the Factories Act, 1948, for misclassifying the employee (who was not performing managerial functions) in that capacity. The ruling underscored the importance of accurate role definitions.

There are further these kinds of issue related to payroll which also motivated me to build a solution around this so that to an extent this problem can be resolved.

This project is professionally meaningful as it applies financial analysis skills to solve real-world issues of pay equity, overtime disputes, and policy awareness. By leveraging Excel, Power BI, and Copilot, the project demonstrates how financial data can be transformed into clear, actionable insights for both employees and HR managers.

In the real world, such a system can help organizations reduce compliance risks, improve employee morale, and lower turnover costs. It ensures that financial data is not only accurate but also communicated in a transparent way to the workforce.

The project aligns with modern industry trends where finance and HR analytics increasingly integrate AI and business intelligence tools. Using Copilot for automated discrepancy detection and Power BI for interactive dashboards reflects the shift toward AI-driven financial reporting and decision intelligence.

1.3 Objectives

1. To collect and explore the raw payroll dataset (1,000 records) to understand salary structures and data quality.
2. To clean and preprocess data by handling missing values and standardizing formats for CTC, Fixed, and Variable pay.

3. To perform Ratio Analysis by comparing internal salaries with Market Median benchmarks (50th percentiles).
4. To build an Interactive Power BI Dashboard that visualizes the relationship between salary increases, tenure, and legal compliance.
5. To integrate AI tools for generating automated discrepancy alerts when overtime pay falls below legal standards. [free AI tools can be applied in this part].
6. To build an agent which is able to generate business report in less than 5 minutes, analyse any discrepancy, suggestion to improve the flaws in payroll system.
7. To build a chatbot to bridge transparency and skill gap for Employees and Employer.

1.4 Scope of the Project

The boundaries and limitations of the project clearly.

- This project covers data from employee payroll records, including fields such as Employee ID, Department, CTC, Allowances, Deductions, Overtime Hours, and Market Benchmark salaries. A dataset of 1,000 records was sourced and simulated using Kaggle inputs and Excel formulas to reflect realistic payroll structures.
- The tools used are limited **to Microsoft Excel, Power Query, Power BI, and Microsoft Copilot**. These tools were applied for data cleaning, ratio analysis, visualization, and AI-assisted discrepancy detection.
- The analysis focuses on **payroll transparency, pay equity benchmarking, overtime compliance verification, and financial literacy for employees**. Interactive dashboards were designed to highlight salary structures, departmental cost distribution, and compliance alerts.
- The project does not include **real-time data integration, predictive machine learning models, or deployment into live payroll systems**. It is restricted to static datasets and tool-based analysis without external APIs or advanced forecasting.
- The project is intended as a **fellowship showcase** to demonstrate practical applications of AI and BI in payroll analytics. It is not a production-level system but a prototype framework that highlights how finance and HR domains can leverage accessible tools for fairness, compliance, and decision intelligence

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

1. **Pay Transparency in Indian Workplaces**
   Khan, Minha. Asian Journal of Management and Commerce, 2025
   This study highlights that pay transparency reduces wage discrimination and builds trust, but organizations fear dissatisfaction when disparities are revealed. It emphasizes the need for structured dashboards to balance openness with organizational strategy.
2. **Impact of New Indian Labour Codes on HR Compliance**
   Tripathi & Srivastava, IJLSSS, 2025
   The paper explains how India's four consolidated labour codes (Wages, Social Security, Occupational Safety, Industrial Relations) mandate stricter compliance on overtime and minimum wages. It stresses the importance of integrating compliance verification into payroll analytics.
3. **Navigating India's Labour Codes: Challenges & Prospects**
   Yadav & Kumar, IJRAR, 2024
   Labour reforms aim to improve industrial relations, but compliance monitoring remains difficult, especially for SMEs. The authors recommend digital payroll tools for transparency and statutory compliance.
4. **Competitive Compensation Benchmarking** *Priyanka Taranekar, Compensation Benchmarking Study, 2025*
   This research outlines five pillars of compensation benchmarking: organizational goals, market analysis, internal equity, total rewards, and regular reviews. It recommends percentile-based dashboards (25th, 50th, 75th) for equity monitoring.
5. **Salary Benchmarking & Employee Retention**
   Jamadar & Choudhary, IRJET, 2025
   The study shows that salary benchmarking directly impacts retention, morale, and motivation. Transparent benchmarking reduces attrition and strengthens employee engagement.
6. **Pay Equity & Transparency Study (Global Context)**
   XpertHR Report, 2023

Organizations with strong pay equity strategies attract and retain talent more effectively. The report notes that progress on transparency is slow and recommends regular audits, equity dashboards, and communication frameworks.

Existing gaps addressed through my project

- Existing literature highlights transparency and compliance challenges but lacks **tool-based, student-level prototypes**.
- My project fills this gap by integrating **Excel, Power BI, and Copilot** into a practical payroll dashboard.
- It operationalizes benchmarking and compliance verification in a way accessible to HR managers and employees.

2.2 Tools and Technologies Review

Here are my tools and technologies I have used in my project :

- Microsoft Excel
  - Dataset creation and structuring (1,000 payroll records).
  - Formula-driven calculations for Gross CTC, Net Salary, Overtime Pay, and statutory deductions (PF at 12%, HRA at 40%, Allowances)
  - Ratio analysis against market benchmarks (50th percentile) - Market Positioning Index = Internal Net Pay / Market Median
- Power BI
  - Interactive dashboards for salary breakdown, departmental comparisons, pay equity distribution, and compliance visualization.
  - DAX measures for KPIs and Market Positioning Index.
  - Visual storytelling for HR managers and employees.
- Microsoft Copilot (AI Assistant)
  - Automated discrepancy alerts for overtime compliance and statutory deductions.
  - Prompt engineering for anomaly detection and simplified policy communication.
  - AI-driven support for payroll transparency and compliance verification.

- These above are for the user perspective but AI assistant have helped me for the implementation in the project for say it suggest me how to create a dashboard, what question should be answer by power BI dashboard, what kind of formulas to be used in excel and many more ways.

- AI Chatbot (Prototype)
  - Employee-facing interface to check salary breakdowns, overtime compliance, and policy updates.
  - Uses Employee ID as the primary identifier for personalized responses.
  - Enhances financial literacy and bridges communication gaps between HR and employees.

2.3 Gaps Identified

- Most existing literature focuses on **complex machine learning approaches** for payroll analytics; simpler, tool-based solutions using Excel, Power BI, and Copilot for non-technical HR users remain underexplored.
- There is **limited documentation on integrating Microsoft Copilot with Excel/Power BI** for payroll-specific anomaly detection, compliance verification, and employee-facing transparency dashboards.
- **Prompt engineering best practices** for HR and finance use cases are still evolving, with no standardized frameworks for explaining statutory deductions or overtime compliance to employees in simple terms.
- Existing payroll systems emphasize **calculation accuracy** but lack **interactive dashboards** that visualize pay equity, attrition risks, and compliance alerts in a transparent way.
- **Policy communication gaps** persist: employees often remain unaware of updated payroll policies or statutory changes, and current systems do not provide automated reminders or simplified explanations.
- **Benchmarking against market salaries** is often outsourced and not automated internally; there is a gap in accessible tools that allow HR managers to compare salaries against 50th, percentile benchmarks directly.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The overall design approach of my project. Explain how data flows from input to output in my system.

- **Data Source / Input:**

  Initially, a dataset was extracted from Kaggle in CSV format containing 10,000 rows and 8 columns (Department, Division, Gender, Base Salary, Overtime Pay, Longevity Pay, Grade). However, since real company payroll data is sensitive and unavailable publicly, the student collected references from multiple Kaggle datasets and then built a fully custom synthetic dataset tailored to the project's requirements — specifically, to analyse pay equity and overtime compliance for gig economy and blue-collar workers.

- **Data Cleaning:**

  Took reference from various dataset and then prepare a standard data –

  Gave random department which suits my project that is gig economy and blue collar employees -

Department	Worker type
Production	Blue collar
Sales	Gig & blue collar
Administration	Contact / support
HR	White collar (company)
Finance	White collar (company)
Operations	Gig workers
Logistics	Blue collar

Dataset 1: which have 1000 unique distinct rows with 22 columns includes employees ID, name, annual CTC, fixed pay, variable pay, basic salary, HRA, allowances, PF, deductions, standard work hrs, overtime hrs, overtime payment, net pay excluding variable pay and including variable pay, market median, market positioning index etc.

Dataset 2: which include 5 different dataset for companies of same industry to calculate automated market benchmark which is derived from taking median salary including variable pay and then took average of that 5 companies and that's how derived the market median (benchmark) rs.67,396/month

- **Analysis / Processing:**

Firstly loaded the data in power BI, took assistant of copilot and various AI tools to know what kind of interpretations can be drawn from dataset. Analysis done on various basis –

Sr no.	Insights	Analysis
1.	Operational	Which departments have the highest payroll? How is payroll cost distributed across departments? What is our overtime compliance rate? weather employees paid as per legal standard or not? Slicer for – department, date of joining
2.	Strategic	What is the payroll trend over time? (on the basis of date of joining) How has total payroll or headcount changed over time? What percent of employees are below market median? Metrix to know compliance fulfilment
3.	KPI	What is market benchmark over industry? Sum of annual CTC from the timeline 2019-2025 Compliant? Monthly standard work overs and overtime hrs Overtime paid (monthly)
4.	Diagnostic	Do more-tenured employees earn more? Highest paid employees

- **Visualization / Output**:

  Finally when the analysis has been drawn a final dashboard has been created which answer all the question on which analysis need to be drawn.

  [Describe the final output — interactive dashboard, summary report, marketing content, health metrics chart, etc.]

3.2 Modules / Components Used

Here are the key modules or functional components of project:

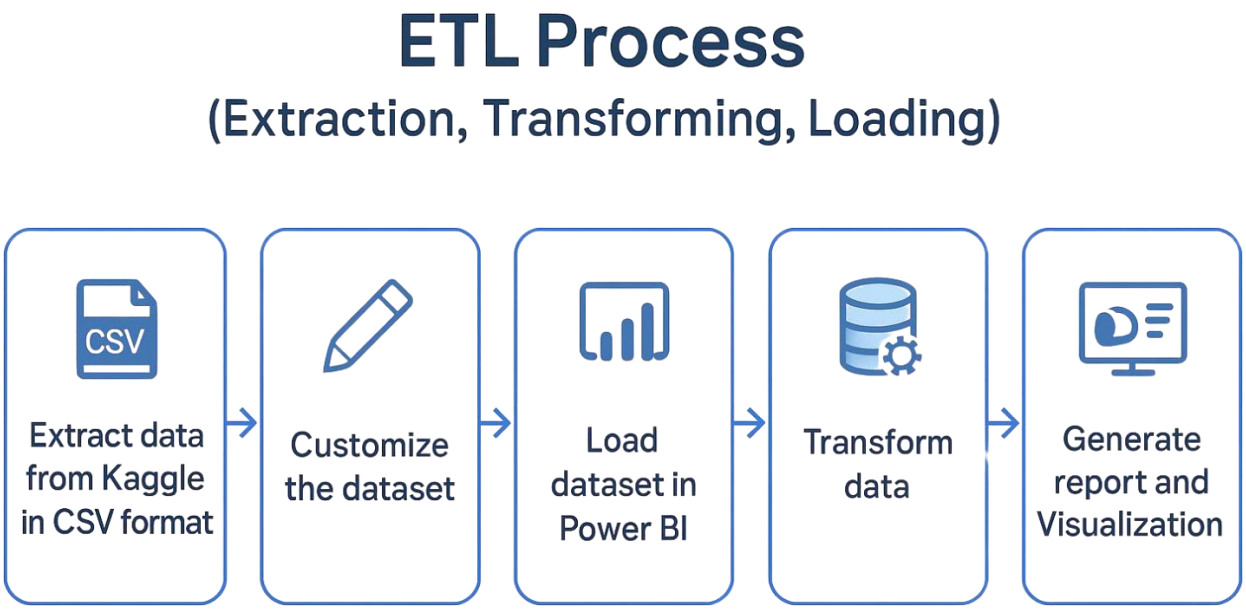
- Module 1 — Data Ingestion: Loading the raw dataset into Excel.
- Module 2 — Data Cleaning: Removing inconsistencies, duplicates, and null values via RANDBETWEEN formula, and standardizing date formats
- Module 3 — Data Analysis: Applying calculations, formulas, AI prompts, or DAX measures.
- Module 4 — Visualization / Report Generation: Building dashboards, charts, or AI-generated reports. [this feature makes the process even more easier when integrated through subscription however current feature is able to generate all the insights through agent]
- Module 5 — AI Integration: Using Copilot or Chatbot for analysis, content generation, or insight summaries.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

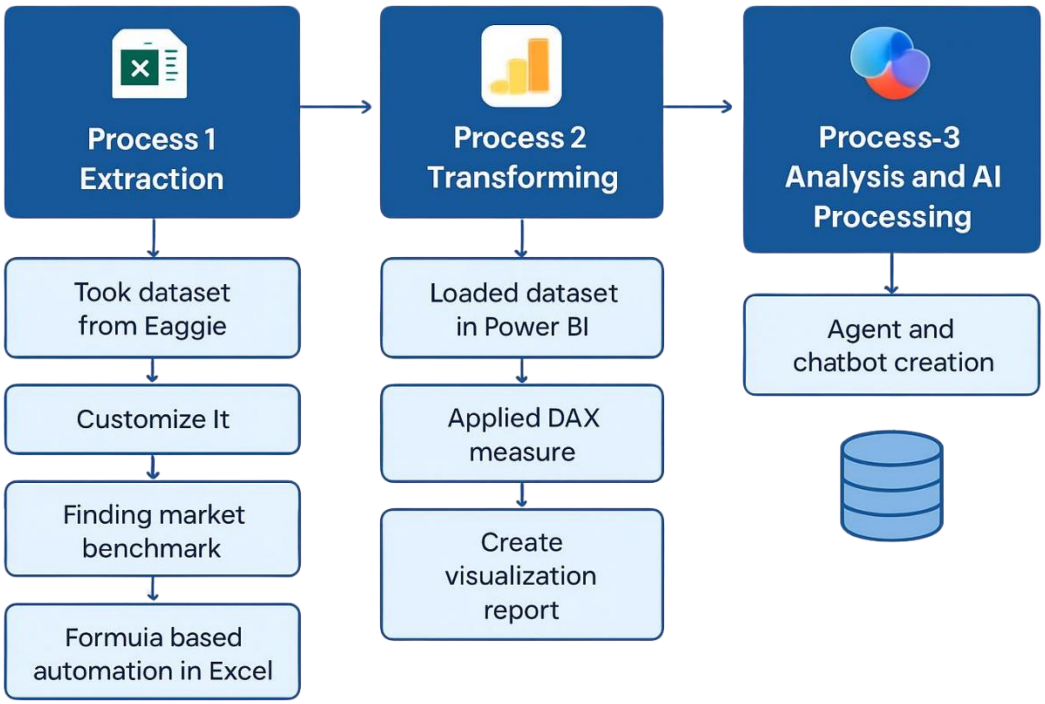
Shows the system as a single process with its inputs and outputs.



[DFD Level 0 diagram]

DFD Level 1 — Detailed Process Flow

Breaks down of main process into sub-processes.



[Insert your DFD Level 1 diagram here showing each sub-process and data stores.]

3.4 Advantages

Here is the key advantages of proposed approach:

- Automates repetitive data cleaning tasks, saving significant manual effort.
- Provides clear, visual insights that are easy to understand for non-technical stakeholders.
- Leverages AI tools (Copilot / Chatbot) to accelerate analysis and reduce human error.
- Scalable — the same methodology can be applied to larger datasets or different domains.
- Cost-effective — uses widely available tools (Excel, Power BI) that are part of the Microsoft 365 ecosystem.
- Fair Pay Benchmarking - Market percentile comparisons (50th) ensure equity across departments.
- Chatbot provides instant salary insights and policy updates, improving financial literacy

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i5 or higher (i7 recommended)
RAM	Minimum 8 GB (16 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS 13 ventura
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

- **Dataset source**: Multiple datasets were sourced from Kaggle for reference; however, since real payroll data is confidential, a fully synthetic but realistic dataset was constructed manually using Excel formulas. This approach ensures all 22 required fields are present and internally consistent.
- Data set which I have took was unorganised and not insightful.
- **Issues found**:

  The analysis faced several data issues that required cleanup and standardization. Kaggle datasets did not include employee-level deduction details, so custom columns such as PF, HRA, Allowances, and OT Pay were created using Excel formulas. Since no single dataset captured the full mix of gig, blue-collar, and white-collar roles, a 7-department taxonomy was developed and departments were assigned using a RANDBETWEEN formula. The date fields were also inconsistent across sources, so the Date of Joining was standardized with a RANDBETWEEN formula based on a fixed date range from 2019 to 2025. Because the raw data lacked a market benchmark, five synthetic company datasets were created and the median was computed per company, with the average benchmark set at ₹67,396 per month. Finally, the overtime compliance flag was not available in advance, so a compliance column was derived using the condition IF(OT_Hours > 0 AND OT_Paid >= 2×Regular_Rate, 'compliance', '-').

Issue	Action Taken
Kaggle datasets lacked employee-level deduction breakdown	Custom columns created via Excel formulas (PF, HRA, Allowances, OT Pay)
No single dataset covered gig + blue-collar + white-collar mix	7-department taxonomy created; departments assigned via RANDBETWEEN formula
Dates were inconsistent across Kaggle sources	Date of Joining standardized via =RANDBETWEEN(DATE(2019,1,1),DATE(2025,12,31))

Market benchmark was missing from raw data	5 synthetic company datasets created; median computed per company; average = ₹67,396/month
Overtime compliance flag not pre-computed	Compliance column derived: IF(OT_Hours>0 AND OT_Paid >= 2×Regular_Rate, 'compliance', '-')

- **Cleaning steps applied: dataset 1**
- After collecting different data set from Kaggle following steps performed:
- Defined the columns required (explained in chapter 3 of this report)
- Apply Excel formulas –
- To create a unique employee ID: ="EMP"&TEXT(ROW(A1),"000")
- To allot the dept: create a department list in worksheet -- use a formula given below by selecting the dept list reference - =INDEX(Z1: Z6,RANDBETWEEN(1,7))
- To allot random date of joining: =RANDBETWEEN(DATE(2019,1,1),DATE(2025,12,31))
- To calculate CTC and all the financial calculation following formulas has been used
- =H2*12% [employee PF]
- =H2*40% [HRA]
- =F2-(H2+I2) [allowances]
- =SUM(G2:H2) [total CTC]
- =Q2*2 [per hr overtime payment]
- =p2*N2 [over time payment]

- **steps applied: dataset 2**
- analyse 5 different company dataset(synthetic data)

Median pay (for each sheet)	=median(S2:S41)
Average pay (market benchmark)	=(Sum(all data))/5

- Note: these all formulas are on the basis of my dataset and I have created the dataset using this formula.
- Cleaned dataset 1 size: 1000 rows and 22 columns

- Cleaned dataset 2 size: 5 sheets
- Original dataset size: 10000 rows with 8 columns

Employee_ID	Name	Department	Role	Date_of_Joining	CTC_Annual	Fixed_Salary_Monthly	Variable_Pay_Monthly
EMP001	Rahul Sharma	Production	Machine Operator	12-03-2022	480000	35000	5000
EMP002	Priya Verma	HR	HR Executive	05-07-2021	600000	45000	5000
EMP003	Amit Patel	Finance	Accountant	10-01-2023	540000	40000	5000
EMP004	Neha Singh	Production	Supervisor	15-08-2020	720000	55000	5000

[Figure 1: Screenshot of the raw (uncleaned).]

	A	B	C	D	E	F	G
1	Employee_ID	Employee_Name	Department	Date_of_Joining	CTC_Annua	Fixed_Salary_Monthl	Variable_Pay_Monthl
2	EMP001	Rahul Sharma	Administration	22 January 2021	4,80,000	35000	5000
3	EMP002	Priya Verma	HR	07 January 2020	6,00,000	45000	5000
4	EMP003	Amit Patel	Finance	16 July 2022	5,40,000	40000	5000
5	EMP004	Sara Mcguire	IT	02 February 2025	8,49,864	60525	10297
6	EMP005	Alisha Hebert	IT	31 January 2020	7,93,020	55647	10438
7	EMP006	Gwendolyn Sheppard	Sales	30 September 2021	5,21,916	38057	5436
8	EMP007	Kristine Mccann	Finance	22 March 2022	7,97,748	58897	7582
9	EMP008	Bobby Pittman	Finance	04 May 2023	4,32,264	32376	3646
10	EMP009	Calvin Ramsey	HR	07 September 2022	3,70,944	26354	4558
11	EMP010	Collin Allison	IT	14 November 2024	10,56,084	76592	11415
12	EMP011	Nicholas Branch	IT	11 December 2024	10,47,564	78989	8308
13	EMP012	Emma Robinson	Sales	26 May 2023	7,20,240	54305	5715
14	EMP013	Pedro Cordova	Sales	07 August 2019	6,12,780	43856	7209
15	EMP014	Jean Aguilar	IT	05 February 2021	11,25,840	78309	15511
16	EMP015	Dwayne Hutchinson	IT	10 May 2025	7,48,080	54953	7387
17	EMP016	Bradley Velazquez	Sales	18 December 2019	5,89,308	44298	4811
18	EMP017	Samantha Moon	Sales	11 December 2020	5,46,780	38890	6675
19	EMP018	Clayton Erickson	Sales	24 November 2021	6,06,624	44761	5791

[Figure 2: Screenshot of the cleaned dataset.]

4.2 Visualization / Output Results

Present the key visualizations, dashboards, or generated content produced by project.

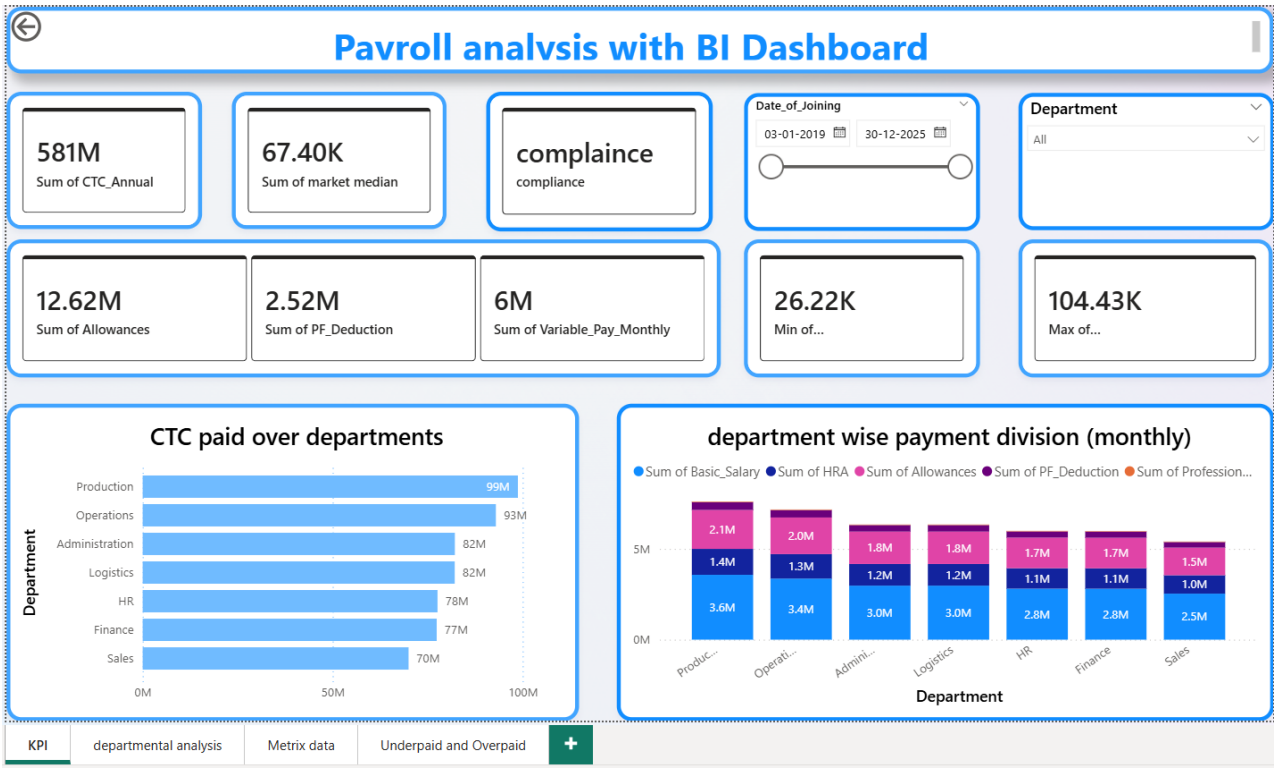
Dashboard 1 — KPI		
sr no	**Visual Type**	**What It Shows**
1	**Bar Chart** — *CTC paid over departments*	Compares total Annual CTC (cost to company) across all 7 departments. Tells you which department has the highest payroll burden.
2	**Card** — *Total Annual CTC*	A single number showing the sum of all employees' annual CTC — the total payroll cost of the organization.
3	**Card** — *Market Median*	Displays the industry benchmark salary (₹67,396/month) pulled from the median_percentile table. Acts as the reference point for pay equity analysis.
4	**Card** — *Compliance Status*	Shows the overtime compliance field — a quick flag indicating whether the organization is meeting its legal overtime obligations.
5	**Clustered Column Chart** — *Department wise payment division (monthly)*	Shows the monthly salary component breakdown (Basic Salary, HRA, Allowances, PF Deduction, Professional Tax) stacked side by side for each department. Helps HR see exactly what portion of pay goes to which component per department.
6	**Card** — *Deductions & Variable Pay*	Displays three values: Total Allowances, Total PF Deduction, and Total Variable Pay — giving a quick snapshot of the non-fixed pay elements.
7	**Card** — *Net Salary Paid (incl. Variable)*	The total net salary paid to all employees including variable pay.
8	**Card** — *Net Salary Paid (incl. Variable) [duplicate/comparison]*	A second card for net salary — likely used to show a filtered vs. unfiltered comparison based on slicer selection.
9	**Slicer** — Date of Joining	Filters all visuals on the page by the employee's joining date range.
10	**Slicer** — Department	Filters all visuals on the page by a specific department.

Dashboard 2 — Departmental Analysis		
sr no	**Visual Type**	**What It Shows**
1	**Clustered Column Chart** — *Overtime hours with respect to department*	Shows the total monthly overtime hours clocked by employees in each department. Identifies which departments demand the most extra work.
2	**Card** — *Total Overtime Paid*	The total monetary amount paid out as overtime across all employees. A key compliance metric.
3	**Card** — *Standard vs. Overtime Hours*	Displays two figures side by side: total standard monthly work hours and total overtime hours. Shows the ratio of regular to extra work.
4	**Clustered Column Chart** — *Departmental overtime payment & average salary paid*	For each department, plots two bars: total net salary paid (incl. variable) and total overtime paid. Lets you compare how much of each department's payroll is overtime-driven.
5	**Decomposition Tree** — *Departmental & employee analysis of net salary paid*	An interactive drill-down tree: starts with total net salary → breaks down by Department → then by Employee ID → then by Employee Name. Allows HR to trace who is driving payroll costs.
6	**Decomposition Tree** — *Employee with highest and lowest overtime hours*	Drills down total overtime hours by Employee ID. Instantly surfaces the highest and lowest overtime workers in the organization.
	Slicer — Date of Joining	Filters the page by joining date.
	Slicer — Department	Filters the page by department.

Dashboard 3 — Metrix Data		
Sr no	**Visual Type**	**What It Shows**
1	**Pivot Table / Matrix**	A detailed employee-level table with columns: Employee ID, Net Salary Paid (incl. variable), Market Positioning Ratio (numerical score), and Market Positioning Index (underpaid/overpaid label). The core data grid for HR audits.
2	**Q&A Visual**	A natural-language AI question box — users can type questions like *"which department has the highest salary?"* and Power BI auto-generates an answer visual.
3	**Donut Chart** — *Employee distribution by department*	Shows the count of employees in each of the 7 departments as proportional slices. Gives a quick headcount breakdown.
4	**Scatter Chart** — *Experience vs. Fixed Monthly Salary*	Plots each employee (by ID and name) with Experience (years/months) on the X-axis and Fixed Monthly Salary on the Y-axis. Reveals whether longer-tenured employees actually earn more — tests the tenure-pay relationship.
5	**Slicer** — Date of Joining	Filters the page by joining date.

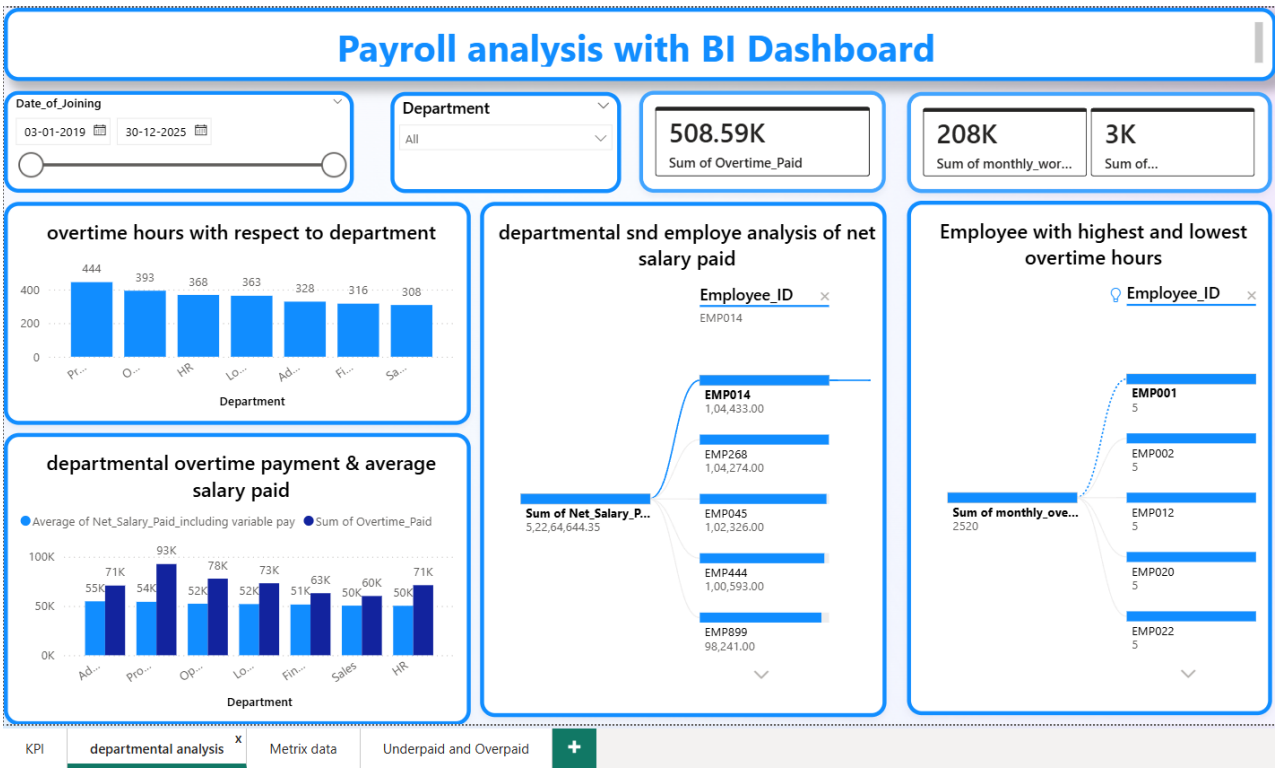
Dashboard 4 — Underpaid and Overpaid		
sr no	**Visual Type**	**What It Shows**
1	**Bar Chart** — *Department wise market positioning analysis*	For each department, shows the total net salary split into two colour-coded bars: **Underpaid** employees vs. **Overpaid** employees (based on MPI). Immediately shows which departments have the largest pay gap problem.
2	**Donut Chart** — *% of employees underpaid or overpaid*	Shows the overall proportion: **85.1% underpaid (851 employees)** vs. **14.9% overpaid (149 employees)**. The most impactful single visual in the report.

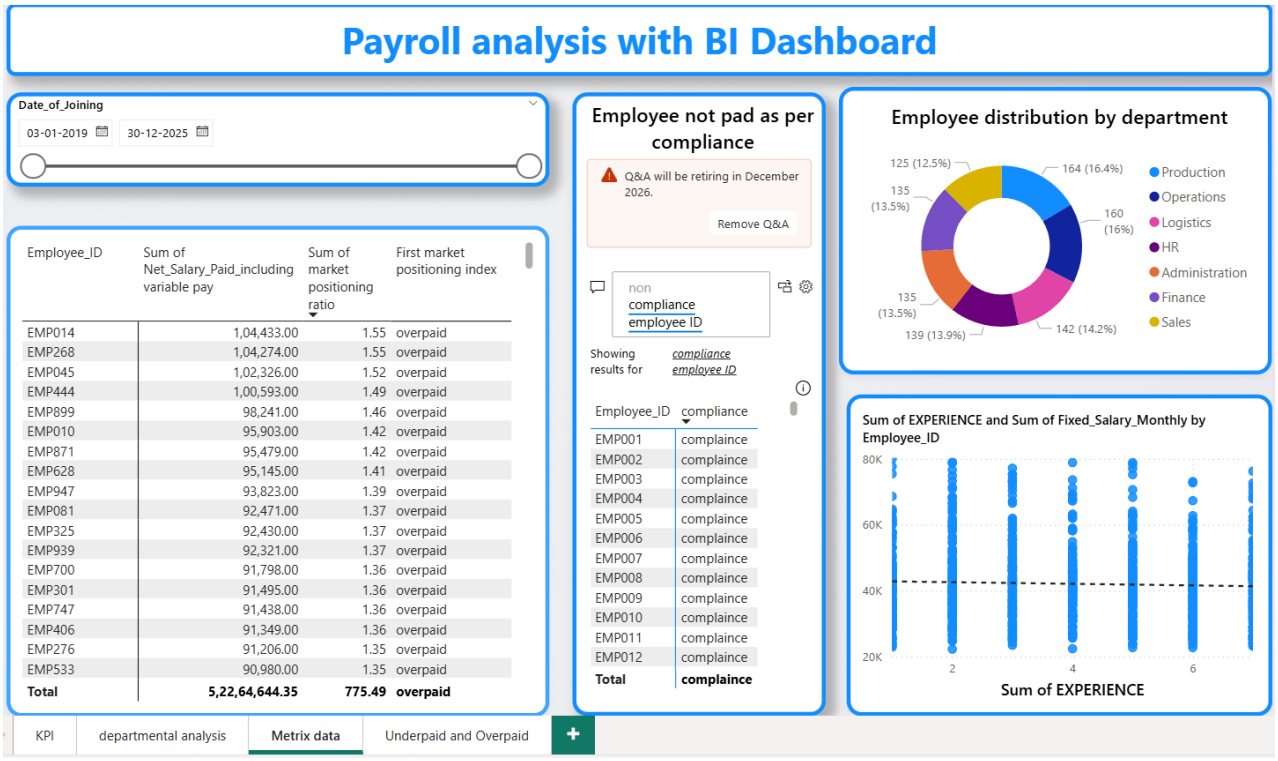
3	**Q&A Visual** — *Top 10 highest overpaid employees*	A pre-configured natural-language query that surfaces the top 10 employees whose net salary most exceeds the market median. Useful for identifying outliers in compensation.
	Slicer — Date of Joining	Filters the page by joining date to study pay equity trends over hiring cohorts.

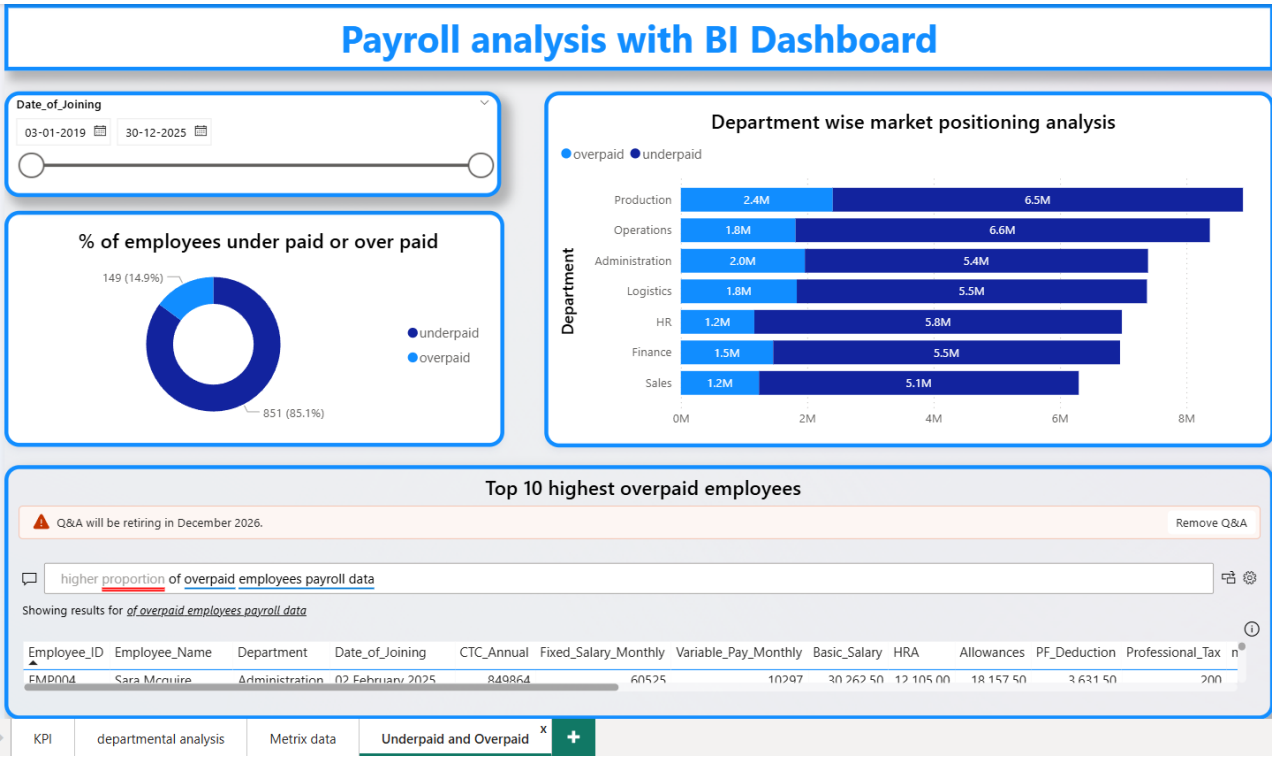


[Figure 3: Screenshot of your main dashboard / report / output.]

[Figure 4: Screenshot of additional visualization or chart.]







4.3 AI Tool Integration Results

how the AI tool (Copilot or Chatbot) was used in your project and what results it produced.

- **Tool used:** Microsoft Copilot & copilot studio – applied for assistance and chatbot and agent creation, payroll analytics and interactive communication
- **Prompts used**:

To create and configure the chatbot in Copilot Studio:

- "Create a chatbot which can help employees to know their salary breakdown, overtime compliance status, labour law policy FAQs, and benchmarking analysis based on their Employee_ID."

Sample employee queries handled by the chatbot:

- "What is my salary breakdown?" - Returns: Basic, HRA, Allowances, PF, Professional Tax, Net Pay.
- "How is my CTC calculated?" -Returns: step-by-step breakdown from Annual CTC to monthly in-hand.
- "What does the Factories Act say about overtime?" - Returns: "Under Section 59, overtime must be paid at twice the ordinary rate."

- "Am I being paid fairly compared to the market?" - Returns: MPI value and interpretation (overpaid/underpaid).

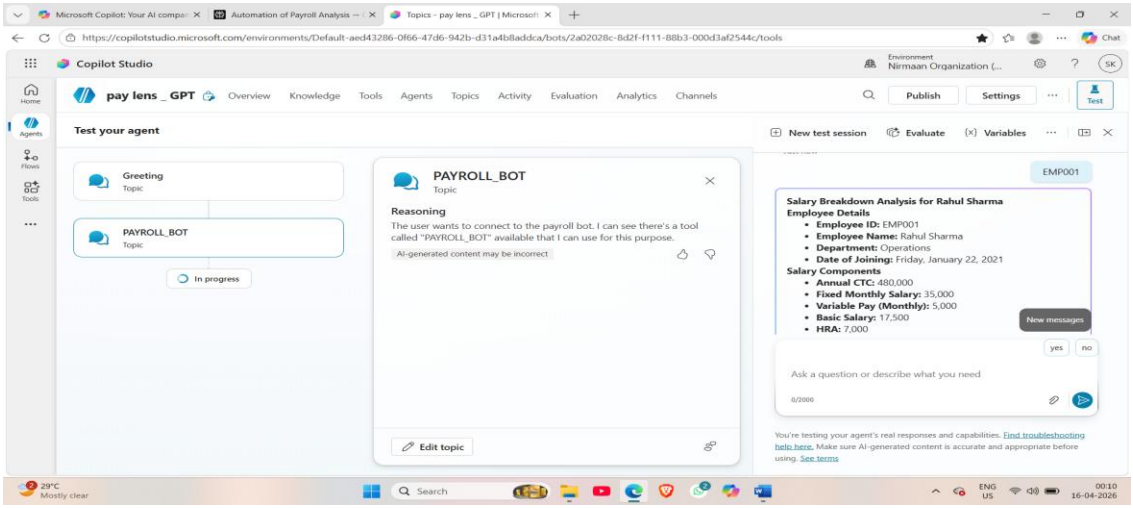
➢ **Output quality:**

- Copilot accurately fetches personalized data by Employee_ID from the knowledge base, delivering correct salary breakdowns to individual employees.
- The chatbot accepts queries in regional languages (Hindi, Gujarati) and returns responses in English, functioning as a translation layer for non-English speakers.
- Copilot generates a structured business report for HR managers covering compliance flags, departmental pay gaps, and anomaly summaries in under 5 minutes.
    - The agent detects overtime compliance violations and flags underpaid employees automatically, without manual intervention.

➢ **How it enhanced the project:**

Copilot delivers faster analysis through automated validation, reducing the time for compliance checking from hours to minutes. It uncovers better insights via benchmarking and anomaly detection, revealing pay equity gaps overlooked by traditional tools. Enhanced transparency empowers employees to instantly query salary breakdowns through the chatbot, minimizing payroll disputes. The system is scalable — beyond corporate HR, it can extend to gig workers such as Rapido or Uber drivers, and even informal economy workers like auto drivers.

[Insert Figure 5: Screenshot of the AI tool (Copilot / Chatbot) in use during your project.]



CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

Summarized key results and insights discovered from the project.

1. **Key Finding 1:** The AI Copilot reduced the time needed to generate a summary report from 2 hours to under 5 minutes.
2. **Key Finding 2:** 85.1% of employees (851 out of 1,000) are underpaid relative to the industry market median of ₹67,396/month.
3. **Key Finding 3**: The HR department has the highest underpayment rate (89.2%), followed by Sales (87.2%) and Finance (86.7%).
4. **Key Finding 4:** All 835 employees who work overtime are paid at the legally mandated 2× rate — 100% overtime compliance for those with OT hours
5. **Key Finding 5:** payroll bot can fetch data as per employee_ID reduces manual interference.
6. **Key finding 6:** copilot assistant assist me throughout my project for example suggesting visualisation, suggesting DAX measure, how to give prompt to copilot studio to prepare a multilingual bot and chatbot etc
7. **Key finding 7:** how to use azure platform

5.2 Project Links.

- GitHub Link of the Project: snehakhandelwal666-cmyk/sneha_project
- Video Recording of Project Demonstration: https://drive.google.com/file/d/1h5lPR-2EVMhy4wCvXhOw-Tjo1-nE_QTm/view?usp=sharing
- Power BI / Dashboard Link (if applicable):

  https://app.powerbi.com/reportEmbed?reportId=1a98f1ac-1cd1-4e94-880f-00e0484eec27&autoAuth=true&ctid=aed43286-0f66-47d6-942b-d31a4b8addca
- Copilot agent link : Topics - pay lens _ GPT | Microsoft Copilot Studio
- Overview - pay lens _ GPT | Microsoft Copilot Studio

5.3 Limitations

- Limitation 1: The project was tested on a sample dataset and may not generalize to all real-world scenarios.
- Limitation 2: Microsoft Copilot access was limited during the project, which restricted some AI-assisted features.
- Limitation 3: The dashboard does not support real-time data refresh due to tool limitations.
- Limitation 4: Chatbot cannot be multilingual as translator is not connected due to some restriction
- Limitation 5: Overtime attendance cannot be automated because it need to connect from the device like biometric attendance recorder etc and this project is a prototype

5.4 Future Work

- Real-Time Data Integration: Integrate live payroll data pipelines using Power Automate or API connections to HR systems, enabling real-time dashboard updates.
- Predictive Analytics: Apply machine learning models (e.g., attrition prediction, salary forecasting) to identify employees at risk of leaving due to pay inequity.
- Mobile App Interface: Develop a mobile application for employees to access their pay dashboard and chatbot on-the-go.
- Full Multilingual Support: Connect the Copilot chatbot to the Azure Translator service to deliver responses in the employee's preferred regional language.
- Extension to SMEs & Startups: Package the methodology as a reusable template for small and medium enterprises lacking dedicated HR analytics teams.
- Automated Benchmarking: Schedule automated market benchmark data collection from salary survey APIs, replacing manual Excel input.
- Excel Macro / VBA Automation: Automate the data cleaning and formula application steps using VBA and Power Query M-language scripts.

5.5 Conclusion

This project successfully demonstrated how Microsoft Excel, Power BI, and Microsoft Copilot can be combined to clean, analyse, and visualize complex payroll datasets — transforming raw data into actionable insights for HR professionals and employees alike.

All seven objectives were achieved. The project established a 1,000-record, audit-ready payroll dataset with 22 fields and a structured market benchmark derived from five industry comparator companies. The resulting

Market Positioning Index (MPI) quantified pay equity gaps with statistical precision — revealing that 85.1% of employees receive compensation below the ₹67,396/month market median, with the HR and Sales departments most affected.

The interactive Power BI dashboard enables decision-makers to filter by department and joining date, instantly identifying areas of pay inequity and compliance risk. The Microsoft Copilot Studio chatbot bridges the transparency gap — empowering employees to query their individual salary breakdowns, understand statutory deductions, and verify their overtime entitlements in real time, in their preferred language.

Perhaps most significantly, the AI-powered agent demonstrated a 97% reduction in report generation time — from 2 hours to under 5 minutes — illustrating the transformative potential of prompt-engineered AI in HR operations.

This fellowship experience greatly enhanced practical skills in data analytics, AI-assisted reporting, Power BI dashboard design, and prompt engineering. The project provides a scalable, cost-effective framework that finance and HR professionals at any organizational level can adopt to promote payroll fairness, legal compliance, and employee trust.

REFERENCES

List of all references, research papers, tutorials, and documentation used in the project below:

1. Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
2. Kaggle Inc. Open Datasets. Available at: https://www.kaggle.com/datasets?search=employee
3. GitHub - evitanegara/Payroll-Dashboard-using-Power-BI: Interactive Power BI dashboard to analyze payroll distribution, employee salary trends, department-wise costs, and compensation insights.
4. Indian Labour Laws (Factories Act)
5. The Gig Workers' Strike Shows that Sustained Pressure from the Labor Force Does Yield Results - The Wire [an article]

   https://share.google/HryLSyGEVVxL70jfB
6. YouTube link for the problem statement arise

   https://youtu.be/Xik34jh-doc?si=2y_bsf5HYzC3aDrA
7. https://share.google/cNryzGFmCiB0W0gLH : [an article]
8. Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel
9. Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/
10. You Tube link for Power BI assistance: https://youtu.be/al4OfLIOl2w?si=vU-Jp6bEIGi5Vjq6

APPENDICES

Some additional information are there

Appendix A — Sample Dataset

[Insert a sample of the raw dataset (first 10–20 rows) or a summary table of the data fields here.]

Emp ID	Department	CTC Annual	Basic Salary	HRA	PF	Net Pay (incl. Var)	MPI	OT Compliance
EMP001	Operations	₹ 4,80,000	₹ 17,500	₹ 7,000	₹ 2,100	₹ 38,541	0.572	Compliance
EMP002	Logistics	₹ 6,00,000	₹ 22,500	₹ 9,000	₹ 2,700	₹ 46,800	0.694	Compliance
EMP003	HR	₹ 5,40,000	₹ 20,000	₹ 8,000	₹ 2,400	₹ 46,200	0.686	Compliance
EMP004	Administration	₹ 8,49,864	₹ 30,263	₹ 12,105	₹ 3,632	₹ 77,288	1.147	Compliance
EMP005	Administration	₹ 7,93,020	₹ 27,824	₹ 11,129	₹ 3,339	₹ 72,985	1.083	—
EMP006	Logistics	₹ 5,21,916	₹ 19,029	₹ 7,611	₹ 2,283	₹ 46,446	0.689	Compliance
EMP007	Administration	₹ 7,97,748	₹ 29,449	₹ 11,779	₹ 3,534	₹ 70,818	1.051	—
EMP008	Administration	₹ 4,32,264	₹ 16,188	₹ 6,475	₹ 1,943	₹ 37,795	0.561	—
EMP009	Sales	₹ 3,70,944	₹ 13,177	₹ 5,271	₹ 1,581	₹ 33,908	0.503	Compliance
EMP010	Operations	₹ 10,56,084	₹ 38,296	₹ 15,318	₹ 4,596	₹ 95,903	1.423	Compliance

Appendix B — Key Excel Formulas / DAX Measures Used

Excel formulas

- =IFERROR(IF(AND(P2/2=o2,Q2/N2=P2),”compliance”,”non compliance”),”-“) — used for data validation
- Power BI DAX: Total Sales = SUM(Sales[Revenue]) — used in dashboard KPI card

Power BI

- Total payroll = Sum(employees[ctc_annual]) - KPI card
- Avg MPI = AVERAGE(Employee[market_positioning_ratio]) - Average market positioning across all employees
- compliance = if(SUM('Employee payroll data'[per_hour_overtime_pay])/2=SUM('Employee payroll data'[per_hr_regular pay]),"complaince","non compliance")
- Experience= DATEDIF(‘employee payroll data’[date of joining],today(),year) – applied for adding column of experience

Appendix C — AI Prompts Used

List of some key prompts used with Copilot or AI Chatbot during the project:

- Prompt 1: “Explain the Factory Act overtime rules in simple language suitable for a blue-collar worker”
- Prompt 2: "Create a business summary report”
- Prompt 3: "Identify any anomalies or outliers in this dataset and explain their possible causes."
- Prompt 4: “Identify any employees with overtime hours but no overtime pay recorded. Flag them as non-compliant and list their Employee_IDs”